

Arduino-based RF Controlled Robot



ROBIN CHALANA

Here we present a simple Arduino-board based robot that can be driven remotely using an RF remote control. This robot can be built very quickly in a small budget. The RF remote control provides the advantage of a good controlling range (up to 100 metres with proper antennae) besides being omnidirectional.

Circuit description

The block diagram of the robot is shown in Fig. 1. It has two major sections: (a) transmitter and (b) receiver and motor driver. The transmitter circuit (Fig. 2) is built around encoder IC HT12E (IC1), 433MHz RF transmitter module (TX1) and a few discrete components. The receiver and motor driver circuit (Fig. 3) is built around Arduino UNO board (BOARD1), decoder IC HT12D (IC2), 433MHz RF receiver module (RX1), motor driver IC L293D (IC3), regulator IC 7805 (IC4) and a few discrete components.

Arduino UNO board. The heart of the robot is Arduino UNO board. Arduino is an Open Source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended for artists, designers, hobbyists, and anyone interested in creating interactive objects or environments.

The Arduino Uno board is based on the ATmega328 microcontroller. It consists of 14 digital input/output pins, six analogue inputs, a USB connection for programming the on-board microcontroller, a power jack, an ICSP header and a reset button. It is operated with a 16MHz crystal oscil-

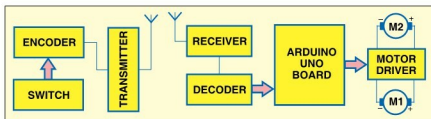


Fig. 1: Block diagram of Arduino-based RF controlled robot

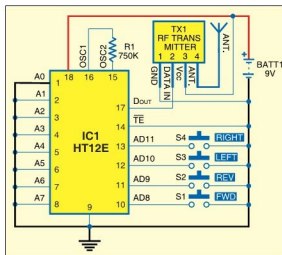


Fig. 2: Circuit of transmitter section

Technical Specifications of Arduino UNO Board

Details	Specifications
Microcontroller	ATmega328
Operating voltage	5V
Input voltage—recommended	7-12V
Input voltage—limits	6-20V
Digital I/O pins	14 (of which six provide PWM output)
Analogue input pins	6
DC current per I/O pin	40 mA
DC current for 3.3V pin	50 mA
Flash memory	32 kB (ATmega328), of which 0.5 kB is used by bootloader
SRAM	2 kB (ATmega328)
EEPROM	1 kB (ATmega328)
Clock speed	16 MHz

lator. It contains everything needed to support the microcontroller. It is very user-friendly; simply connect

it to a computer with a USB cable to get started. The microcontroller on the board is programmed using the Arduino programming language and the Arduino development environment.

Remote control. For controlling the robot remotely, Holtek's encoder-decoder pair (HT12E and HT12D) together with a 433MHz transmitter-receiver pair is used.

HT12E and HT12D are CMOS ICs with working voltage ranging from 2.4V to 12V. Encoder HT12E has eight address and another four address/data lines. The data set on these twelve lines (address and address/data lines) is serially transmitted when transmit-enable pin TE is taken low. The data output appears serially on D_{OUT} pin.

The data is transmitted four times in succession. It consists of differing lengths of positive-going pulses for '1' and '0', the pulse-width for '0' being twice the pulse-width for '1'. The frequency of these

pulses may lie between 1.5 and 7 kHz depending on the resistor value between OSC1 and OSC2 pins.